

Kewalin Samart

PhD Candidate in Computational Bioscience

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Education

University of Colorado Denver, Anschutz Medical Campus (CU Anschutz)

Aurora, CO

Doctor of Philosophy in Computational Bioscience

Aug 2022 – present

- Thesis Advisors: Dr. Janani Ravi and Dr. Arjun Krishnan
- Thesis Committee:
Dr. Antonio Porras (chair), Dr. Christina Stallings, Dr. Milton Pividori, Dr. Natalie Davidson
- Hirs Fellowship
- Coursework: Methods and Tools in Biomedical Informatics, Research Methods in Biomedical Informatics, Machine Learning for Biomedical Applications, Statistics for Basic Sciences, Neural Networks and Deep Learning, Rethinking Data Analysis

Michigan State University (MSU)

East Lansing, MI

Bachelor of Science in Computational Mathematics

Degree Awarded May 2022

Minor in Computational Math, Science, and Engineering

- High Honor (Summa Cum Laude)
- McCartney Math Award, L.C. Plant Math Award, College of Natural Science Scholar
- Cumulative GPA: 3.98/4.00; Major GPA: 4.00/4.00
- Dean's Honor List: Fall 2018 – Spring 2022
- Royal Thai Government Scholarship Recipient 2017

Skills

- **Programming languages:** R, Python, Shell, C++, TeX
- **Computational Tools:** Git, vi, Terminal, Unix/Linux, High-performance/ Cluster computing, LaTeX

Publications

- **Kewalin Samart**^{**}, Phoebe Tuyishime^{**}, Arjun Krishnan^{*}, Janani Ravi^{*}. Reconciling multiple connectivity scores for drug repurposing. *Briefings in Bioinformatics* (2021)
doi.org/10.1093/bib/bbab161. [GitHub: https://github.com/jravidlab/connectivity_scores](https://github.com/jravidlab/connectivity_scores)
[Live doc: https://jravidlab.github.io/connectivity_scores](https://jravidlab.github.io/connectivity_scores)
- Janani Ravi^{*}, **Kewalin Samart**, Jason Zwolak. Modeling the START transition in the budding yeast cell cycle. *PLOS Computational Biology* (2024) doi.org/10.1371/journal.pcbi.1012048. [Online simulator: http://sbmlsimulator.org/simulator/by-start](http://sbmlsimulator.org/simulator/by-start)

Preprints

- **Kewalin Samart**, Ling Thang, Landon Buskirk, Amy Tonielli, Arjun Krishnan^{*}, Janani Ravi^{*}. Integrative transcriptome-based drug repurposing in tuberculosis. *bioRxiv* (2025).
doi.org/10.1101/2025.06.02.657296
- Hao Yuan, Aishwarya Mandava, **Kewalin Samart**, Julia Ganz, Arjun Krishnan^{*}. Linking genetic risk to disease-relevant cellular states via metacell-informed modeling with ICePop. *bioRxiv* (2026).
doi.org/10.64898/2026.04.01.715877

^{**}co-primary authors; ^{*}corresponding authors

Conferences, meetings, workshops, seminars

Invited Talks | Extramural

- Sydney Statistical Bioinformatics Seminar. Sydney, Australia, Apr 20, 2026. Virtual. *Integrative transcriptome-based drug repurposing in tuberculosis*. [Recording](#)

Contributed Talks | Extramural

- 2025 Great Lakes Bioinformatics Conference (GLBIO), the University of Minnesota, Minneapolis, MN, May 2025. Contributed Talk. *SigMatch: a transcriptome-based regression method to detect mechanistic links across diseases and drugs*.
- 2024 Quantitative Cell and Molecular Biology (qCMB) Symposium. Colorado State University, Fort Collins, CO, Jun. 18, 2024. Contributed Talk. *Leveraging transcriptomic disease- and drug signatures towards identifying novel host-directed therapeutics for tuberculosis*.
- R workshop at Global Alliance for Rapid Diagnostics (GARD). Virtual, Mar. 21, 2023. Presentation material: https://github.com/jananiravi/R_intro. *Introduction to R programming*.
- 2nd Latin American Congress of Women in Bioinformatics and Data Science (WBDS LA). Virtual, Sep. 23, 2021. Contributed Talk. *Reconciling Multiple Connectivity Scores for Drug Repurposing*.
- 29th Conference on Intelligent Systems for Molecular Biology and the 20th European Conference on Computational Biology (ISMB/ECCB). Virtual, Jul. 25, 2021. Contributed Talk. *Reconciling Multiple Connectivity Scores for Drug Repurposing*.
- R-Ladies Pune. Virtual, May. 9, 2021. Presentation material: <https://github.com/KewalinSamart/Rladies-lightning-talk-distill>. Lighting Talk. *Distill-flavoured R Markdown for Scientific Manuscripts*.
- *Repurposing drugs for diseases*. The Sci-Files on Impact 89FM Episode 11 Season 6, Aug. 1, 2021. Podcast: <https://impact89fm.org/101365/podcasts/the-sci-files-08-01-2021-kewalin-samart-repurposing-drugs-for-diseases/>

Contributed Talks | Intramural

- 2025 CU Anschutz Three-minute Thesis (3MT) Competition. University of Colorado Anschutz Medical Campus, Aurora, CO. Dec. 08, 2025. *Drug repurposing for tuberculosis (TB): Learning from modern diseases*. [Recording](#)
- Computational Bioscience Program Seminars. CU Anschutz, Aurora, CO. Dec. 3, 2025. Research Updates. *SigMatch: a regression-based approach to detect mechanistic links across diseases and drugs*.
- 2025 CU Department of Biomedical Informatics Annual Retreat. University of Colorado Anschutz Medical Campus, Aurora, CO. Aug. 26, 2025. Lightning Talk. *SigMatch: a regression-based approach to detect mechanistic links across diseases and drugs*.
- R-Ladies Aurora Spring 2025 Workshop Series. CU Anschutz, Aurora, CO. Feb. 18, 2025. 1.5-hours hands-on workshop. Presentation materials: <https://github.com/rladies-aurora/202502-Python-in-R>. *Python in R*.
- 2024 CU Department of Biomedical Informatics Annual Retreat. Denver Botanical Gardens, Denver, CO. Aug. 27, 2024. Lightning Talk. *SigMatch: Mechanism-guided drug repurposing in infectious diseases*.
- Computational Bioscience Program Seminars. CU Anschutz, Aurora, CO. Jan. 22, 2025. Comprehensive Exam. *Unraveling disease-specific signatures towards identifying host-directed therapeutics for infectious diseases*.
- Computational Bioscience Program Seminars. CU Anschutz, Aurora, CO. May. 6, 2024. Research Updates. *Unraveling disease-specific signatures towards identifying host-directed therapeutics for infectious diseases*.

Selected Posters | Extramural

- 34th Conference on Intelligent Systems for Molecular Biology (ISMB). Washington, DC, July 12–16, 2026. *SigMatch: a transcriptome-based regression method for cross-disease drug repurposing*.
- 2024 Computational Systems for Integrative Genomics. Flatiron Institute, New York, NY, Jul. 17–18, 2024. *Leveraging transcriptomic disease- and drug signatures towards identifying novel host-directed therapeutics for tuberculosis*.
- 2023 Genome Informatics Conference. Cold Spring Harbor Laboratory, Woodbury, NY, Dec. 6–9, 2023. *Integrating multiple transcriptome-based methods to repurpose drugs for infectious diseases*.
- Intelligent Systems for Molecular Biology, ISMB 2022, International Society for Computational Biology flagship conference. Virtual, Jul. 11–12, 2022. *Reconciling Multiple Connectivity Scores for Drug Repurposing*.
- 2nd Latin American Congress of Women in Bioinformatics and Data Science (WBDS LA). Virtual, Sep. 22 – 24, 2021. *Reconciling Multiple Connectivity Scores for Drug Repurposing*.
- 29th Conference on Intelligent Systems for Molecular Biology and the 20th European Conference on Computational Biology (ISMB/ECCB). Virtual, Jul. 25–30, 2021. *Reconciling Multiple Connectivity Scores for Drug Repurposing*.
- 2021 Great Lakes Bioinformatics Conference (GLBIO). Virtual, May 10–13, 2021. *Reconciling Multiple Connectivity Scores for Drug Repurposing*.

Selected Posters | Intramural

- 2025 CU Department of Biomedical Informatics Annual Retreat. Aurora, CO. Aug. 26, 2025. *SigMatch: a regression-based approach to detect mechanistic links across diseases and drugs*.
- 2025 Computational Bioscience Program Annual Retreat. Boettcher Mansion-Lookout Mt. Golden, CO. May. 1, 2025. *SigMatch: a regression-based approach to detect mechanistic links across diseases and drugs*. [Best poster]
- Colorado Immunology and Microbiology conference. Vail, CO, Sep. 24–26, 2022. *Integrating multiple transcriptome-based methods to repurpose drugs for infectious diseases*.

Peer reviews

Peer-reviewed computational drug repurposing manuscripts for *PLOS ONE* and *Bioinformatics*.

Fellowships, honors, awards

- Mar 2025 Travel Fellowship
Great Lakes Bioinformatics Conference 2025, Minneapolis, Minnesota, USA
- Aug 2022 Hirs Fellowship
University of Colorado Denver, Graduate School
- Mar 2022 Rasmussen Graduate Recruitment Award
Michigan State University (*declined award*)
- Mar 2022 College of Engineering Distinguished Scholarship for Graduate Studies
Michigan State University (*declined award*)
- Mar 2022 The Graham and Braddock awards
Pennsylvania State University (*declined award*)
- May 2022 High Honor (Summa Cum Laude)
Michigan State University
- Apr 2022 McCartney Mathematics Award
Department of Mathematics, Michigan State University
- Oct 2021 Fall 2021 CNS Undergraduate Research Scholarship
College of Natural Science, Michigan State University

- May 2021 Summer Research Scholarship, EnSURE program
College of Engineering, Michigan State University
- Jan 2021 Spring 2021 CNS Undergraduate Research Scholarship
College of Natural Science, Michigan State University
- Oct 2020 Fall 2020 CNS Undergraduate Research Scholarship
College of Natural Science, Michigan State University
- May 2020 Summer 2020 CNS Undergraduate Research Scholarship
College of Natural Science, Michigan State University
- Apr 2020 L.C. Plant Mathematics Award Scholarship
Department of Mathematics, Michigan State University
- Jan 2020 Spring 2020 CNS Undergraduate Research Scholarship
College of Natural Science, Michigan State University
- Mar 2018 Academic Excellence Award (High Honor)
Lawrence Academy, Groton, MA
- Dec 2017 Academic Excellence Award (Honor)
Lawrence Academy, Groton, MA
- May 2017 Full-ride Scholarship for undergraduate and graduate studies in the U.S.
National Center for Genetic Engineering and Biotechnology (BIOTEC),
Royal Thai Government, Bangkok, Thailand

Research experience

Graduate Research Assistant

Jul 2023–

JRavi and Krishnan Labs; PIs: Dr. Janani Ravi and Dr. Arjun Krishnan

Department of Biomedical Informatics, CU Anschutz

- Studying infectious diseases with a focus on host-directed drug repurposing

PhD Rotation Student

Mar 2023–May 2023

Zhang Lab; PI: Dr. Fan Zhang

Department of Biomedical Informatics, CU Anschutz

- Predicting drug response pathway-driven cell phenotypes in Rheumatoid arthritis using large single-cell transcriptomics.

PhD Rotation Student

Jan 2023–Mar 2023

Kechris Lab; PI: Dr. Katerina Kechris

Department of Biostatistics and Informatics, CU Anschutz

- Identification of enriched pathways associated with multiple Chronic Obstructive Pulmonary Disease phenotypes.

Undergraduate Research Assistant

Mar 2021–May 2022

Krishnan Lab; PI Dr. Arjun Krishnan

Department of Computational Mathematics, Science, and Engineering, MSU

- Research focus: methods and applications of transcriptome-based drug repurposing.

Undergraduate Research Assistant

Dec 2019–May 2022

JRavi Lab; PI: Dr. Janani Ravi

Department of Pathobiology and Diagnostic Investigation, MSU

- Research focus: methods and applications of transcriptome-based drug repurposing and mathematical modeling.

Teaching experience

Educational Assistant

Spring 2024

Computational Bioscience Graduate Program, CU Anschutz

CPBS 7712: Research Methods in Biomedical Informatics

Supported student learning by assisting in classes, hosting office hours, and facilitating class discussions to promote engagement and understanding of course material.

Course director: Dr. Anne Thessen

Undergraduate Teaching Assistant

Jan 2022–Apr 2022

Department of Computational Mathematics, Science, and Engineering, MSU

CMSE 202: Computational Modeling and Data Analysis II

Lead instructor: Dr. Luciano Silvestri

Mentoring

High schoolers

- Summer 2025 Asha Masoudi
Phillips Exeter Academy, New Hampshire
- Summer 2025 Nyjah Munn
DSST: Conservatory Green High School, Colorado

Undergraduates

- May 2025– Ling Thang
Computer Science, Metropolitan State University of Denver
- Oct 2023– Landon Buskirk
Data Science, Michigan State University
- Summer 2024 Kritika Verma
IT and Mathematical Innovations, University of Delhi
- Spring 2022 Amy Tonielli
Biomedical Laboratory Science, Michigan State University

PhD rotation students

- Sep 2024–Nov 2024 Michael Opoku Ntrakwa
Human Medical Genetics and Genomics Program, CU Anschutz.
- Dec 2024–Feb 2025 Yichi Chen
Human Genetics and Genomics Program, CU Anschutz.

Professional services

- 2024–2026 Computational Bioscience Program Admissions Committee
- 2023 Computational Bioscience Program Retreat Committee
- 2021–2022 the University Academic Grievance Hearing Board
the University Academic Integrity Hearing Board
the University Academic Appeal Board

Outreach

- June 2025 MONET: Multi Omics NETwork Analysis Workshop
Teaching assistant for module 1: Introduction to Network Science and NHGRI/AnVIL system
- Aug 2023– Co-organizer of R-Ladies Aurora
Plan and organize events for R-Ladies Aurora
(over 230 members on Meetup: meetup.com/rladies-aurora/)

- Jul 2023 2023 Summer Academy, SACNAS CU Anschutz Chapter
Helped set up program activities.
- Jul 2021 useR! 2021 The R Conference
MaRmot. Helped provide program information to participants.
- Mar 2018 The Arc of Opportunity in North Central Massachusetts
Cared for people with disabilities.

Professional membership

- 2021– International Society for Computational Biology (ISCB), Bethesda, MD
- 2021–2025 Association for Women in Mathematics (AWM), Providence, RI
- 2021–2022 MSU Women in STEM, Member East Lansing, MI
- 2020–2022 Women + Data Science, Member East Lansing, MI
- 2019–2022 R-Ladies East Lansing, Member East Lansing, MI